

THEORY OF FLOW LOCALISATION DUE TO DYNAMIC STRAIN AGEING

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Abstract—The localisation of plastic flow due to dynamic strain ageing is evaluated taking into account transient behaviour associated with the time dependence of the solute composition at mobile dislocations. A constitutive flow equation is defined which includes the local solute composition as an implicit state variable. Linear perturbation theory is employed to define an instability strain above which strain localisation occurs. The instability strain is found to be greater than the strain required for a negative steady state strain rate sensitivity of the flow stress. Both linear theory and numerical simulation show that strain localisation is characterised by sinusoidal strain and strain rate oscillations of increasing amplitude, leading to deformation band formation. The strain required for observable strain localisation is shown to depend on specimen uniformity and experimental technique. The results of the analysis are in agreement with experimental measurements in Fe-C and Au-Cu alloys.

Résumé—La localisation de l'écoulement plastique due au vieillissement dynamique est évaluée en tenant compte du comportement transitoire associé à la dépendance temporelle de la concentration en soluté sur les dislocations mobiles. On définit une équation d'état de l'écoulement dans laquelle la composition locale en soluté est une variable d'état implicite. La théorie linéaire des perturbations est utilisée pour définir une déformation d'instabilité au dessus de laquelle la localisation de la déformation se produit. On trouve que la déformation d'instabilité est supérieure à la déformation nécessaire à une sensibilité négative de la contrainte d'écoulement à la vitesse de déformation. La théorie linéaire et les simulations numériques montrent toutes deux que la localisation de la déformation est caractérisée par une déformation sinusoidale ainsi que par des oscillations de la vitesse de déformation d'amplitude croissante, provoquant la formation de bandes de déformation. La déformation nécessaire à l'observation de la localisation dépend de l'uniformité de l'échantillon et de la technique expérimentale. Les résultats de l'analyse s'accordent avec les mesures expérimentales dans les alliages FeC et AuCu.

Zusammenfassung—Die Lokalisierung des plastischen Fließens wegen dynamischer Reckalterung unter Berücksichtigung des transienten Verhaltens, welches mit der Zeitabhängigkeit der Zusammensetzung der Verunreinigungswolke zusammenhängt wird untersucht. Es wird eine Grund-Fließgleichung definiert, die die lokale Zusammensetzung an gelösten Stoffen als eine implizite Zustandsvariable enthält. Mit der linearen Störungstheorie wird eine Instabilitätsdehnung definiert, oberhalb der Dehnungslokalisierung auftritt. Die Instabilitätsdehnung ergibt sich als größer als diejenige Dehnung, die für eine negative Dehnungsratenempfindlichkeit der Fließspannung erforderlich ist. Sowohl lineare Theorie wie auch numerische Simulation zeigen, daß die Dehnungslokalisierung durch sinusförmige Dehnungs- und Dehnungsratenoszillationen charakterisiert ist, welches zur Bildung von Verformungsbändern führt. Die für beobachtbare Dehnungslokalisierung notwendige Dehnung hängt von der Gleichförmigkeit der Proben und der experimentellen Technik ab. Die Ergebnisse der Analyse stimmen mit experimentellen Messungen an Fe-C- und Au-Cu-Legierungen überein.

1. INTRODUCTION

The interaction between solute atoms and mobile dislocations during plastic deformation is well known to result in non-uniform flow. In tensile tests localised yielding is observed to develop on reaching an initial critical strain, ϵ_c and subsequent deformation occurs via the repeated initiation and propagation of deformation bands along the gauge length (Portevin-Le Chatelier effect).

Phenomenological studies [1-6] associate the occurrence of the Portevin-Le Chatelier effect with a negative strain rate dependence of the flow stress. Penning [3] and recently Kubin and Estrin [6] have examined the dynamics of repeated discontinuous

yielding for materials exhibiting a bounded region of negative strain rate sensitivity. Homogeneous deformation is unstable for the region where $d\sigma/d\dot{\epsilon}$ is negative. Whenever this region is reached, whether by increasing strain or stress, discontinuous jumps in strain rate across the unstable region should occur.

Van den Beukel [7] first showed that a negative strain rate dependence of the flow stress can result from the diffusion of solute atoms to dislocations temporarily arrested at obstacles in the slip path. At temperatures and strain rates where solute diffusion to dislocations occurs [1, 8] the local solute composition at arrested dislocations, C_s , is a function of the waiting time at obstacles, τ_w . A decrease in flow stress may then accompany an increase in strain rate

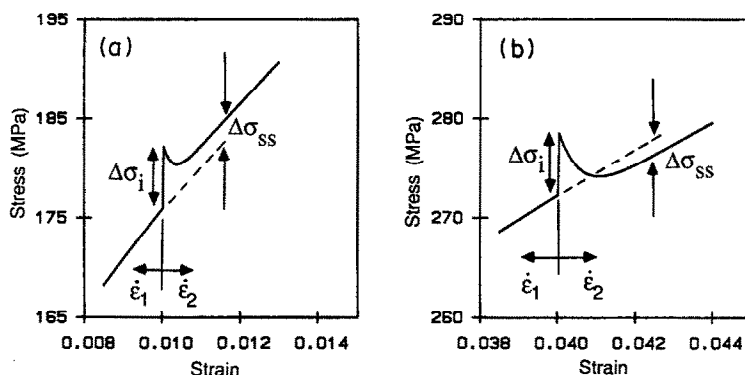


Fig. 1. (a) Calculated flow stress transient associated with a discontinuous increase in strain rate; $\dot{\epsilon}_1 = 4 \times 10^{-5} \text{ s}^{-1}$, $\dot{\epsilon}_2/\dot{\epsilon}_1 = 10$, $\epsilon = 0.01$. (b) Calculated flow stress transient associated with a discontinuous increase in strain rate; $\dot{\epsilon}_1 = 4 \times 10^{-5} \text{ s}^{-1}$, $\dot{\epsilon}_2/\dot{\epsilon}_1 = 10$, $\epsilon = 0.04$.

due to the decrease in C_s resulting from the decrease in t_w . However, the solute composition at dislocations cannot instantaneously respond to a change in strain rate. As a consequence, a transient period may be expected to follow the change in strain rate as C_s adjusts to its new quasi-steady state value (Fig. 1). In the transient region the time dependent variation of the flow stress is determined by the rate of change of the flow stress associated with the time dependent change in C_s and the rate of strain hardening.

Although the actual process by which the new quasi-steady state value of C_s is reached may differ for increases and decreases, respectively in strain rate, the time constant of the transient period should be of the order of t_w of the new strain rate [9]. For example, if the strain rate is abruptly decreased the mobile dislocations must on the average wait for a period of time $t \sim t_w$ before acquiring the higher quasi-steady state composition characteristic of the new strain rate by solute diffusion. For an increase in strain rate it appears that the dislocations on the average must move at least once before quasi-steady conditions are restored.

As shown in Fig. 1, two different strain rate sensitivities may be defined. The instantaneous strain rate sensitivity, $M_i = \Delta\sigma_i/\Delta \ln \dot{\epsilon}$, corresponds to conditions of constant ϵ and C_s . The quasi-steady state strain rate sensitivity, $M_f = \Delta\sigma_{ss}/\Delta \ln \dot{\epsilon}$, includes the effect of the steady state strain rate dependence of C_s , on the change in flow stress. Numerous experimental measurements have shown that M_i remains positive at all strains, while M_f decreases with increasing strain [9–16], becoming negative at a strain, ϵ_0 . The extent of the transient region is observed to increase with increasing strain [9, 17].

Since the Portevin–Le Chatelier effect is observed to initiate only when $M_f < 0$ there is a clear correlation between negative steady state strain rate sensitivity and the initiation of localised flow. However, measurements of ϵ_0 are found to substantially exceed ϵ_0 [9–14], indicating that $M_f = 0$ is not a valid criterion for flow localisation. In this work the crite-

riion for the onset of localised flow is re-examined, taking into account the inherent transient response of the flow stress to changes in strain rate in the region of dynamic strain ageing.

2. CONSTITUTIVE FLOW EQUATION

It is generally accepted that plastic flow resulting from thermally activated dislocation motion may be modelled using an Arrhenius equation of the form

$$\dot{\epsilon} = A \exp(-H/kT) \quad (1)$$

where H is the activation enthalpy for dislocation motion. Assuming a linear dependence of H on σ and the effective local solute composition impeding dislocation motion, C_s and taking account of the Orowan equation ($\dot{\epsilon} = \rho_m b L/t_w$) gives

$$\dot{\epsilon} = A_0 \rho_m \exp[-H'_c C_s] \exp[(\sigma - \sigma_f)/M_i] \quad (2)$$

where ρ_m is the mobile dislocation density, $H'_c = \partial H/\partial C_s$, and σ_f is the internal stress. σ_f is usually taken to be a function of the forest dislocation density, ρ_f , $\sigma_f = \alpha G b \rho_f^{1/2}$, where G is the shear modulus and b is the Burgers vector. M_i is the instantaneous strain rate sensitivity, $M_i = [d\sigma/d \ln \dot{\epsilon}]_{\epsilon, C_s}$ and is positive at all strains. Equation (2) is equivalent to that proposed by Estrin and Kubin [18] but takes into account the effect of C_s on the flow stress. It is assumed that ρ_f , ρ_m and C_s are state variables which describe the average behaviour of individual dislocations.

As pointed out by van den Beukel [7], C_s is not constant in the regime of dynamic strain ageing, but rather will be a function of the flow of solute atoms to arrested mobile dislocations. Assuming Cottrell–Bilby strain ageing kinetics [19] the solute composition may be expressed as

$$C_s = C_0 (K D t_a)^{2/3} \quad (3)$$

where C_0 is the alloy solute composition ($C_s \gg C_0$), D is the solute diffusion coefficient, t_a is the effective time that the arrested mobile dislocations have aged, and K is a constant which includes the solute-

dislocation binding energy. Equation (3) is valid only for the initial stages of ageing. Louat's [20] generalisation of the Cottrell-Bilby analysis to include saturation at longer ageing times gives

$$C_s = C_m \left[1 - \exp\left(-\frac{C_0(KDt_a)^{2/3}}{C_m}\right) \right] \quad (4)$$

where C_m is the saturation value of C_s .

Rewriting equation (2) and combining with equation (3) gives the flow stress equation for constant strain rate as:

$$\sigma = \sigma_f + M_i \left(\ln \frac{\dot{\epsilon}}{\dot{\epsilon}_0} + H'_c C_0 (KDt_a)^{2/3} \right) \quad (5)$$

where $\dot{\epsilon}_0 = A_0 \rho_m$.

2.1. Strain rate sensitivity

Previous treatments of the Portevin-Le Chatelier effect have assumed that the effective ageing time in equations (3) and (4) is equal to t_w . However, as discussed above, C_s cannot instantaneously respond to a change in strain rate. Assuming that the variation in C_s immediately following an abrupt change in strain rate follows first order relaxation kinetics, the strain rate sensitivity may be expressed as [21]

$$M = \frac{d\sigma}{d \ln \dot{\epsilon}} \bigg|_t = M_i + (M_f - M_i) \left[1 - \exp\left(-\frac{t}{\tau}\right) \right] \quad (6)$$

where $M_f = M_i(1 - 2H'_c C_s/3)$ and τ is a relaxation time which is expected to be of the order of t_w . The strain and strain rate dependence of M_f may be expressed as [7], $M_f = M_i[1 - K_1(\epsilon^m + \beta/\dot{\epsilon})^{2/3}]$, where the strain exponents m and β may be associated with the strain dependence of the mobile dislocation density, $\rho_m \propto \epsilon^\beta$ and the strain enhanced vacancy concentration, $C_v \propto \epsilon^m$ ($m = 0$ for interstitial alloys).

Equation (6) shows explicitly the transient behaviour of M associated with the time dependent change in C_s accompanying a strain rate change. The curves shown in Fig. 1 were calculated from equations (5) and (6) for a factor of 10 increase in strain rate using the data in Table 1 for low carbon steel [11]. [For large changes in strain rate (from $\dot{\epsilon}_1$ to $\dot{\epsilon}_2$) the factor 2/3 in the expression for M_f must be replaced by $(1 - \dot{\epsilon}_1/\dot{\epsilon}_2)^{2/3}/\Delta \ln \dot{\epsilon}$.]

The time dependence of the strain rate sensitivity following a strain rate change is consistent with t_a obeying a Boltzmann type relaxation equation.

$$\frac{dt_a}{dt} = -\left(\frac{t_a - t_w}{\tau}\right) \quad (7)$$

Equation (7) gives t_a as a function of the strain and strain rate history of the material, which can be

determined from a hereditary integral if $t_w(t)$ is known. If $t_w \gg t_a$, then from equation (7) $dt_a/dt \sim 1$, in agreement with the fact that the solute concentration at arrested dislocations cannot increase faster than that allowed by the passage of time. On the other hand it may be shown that if t_w is slowly varying the current difference between t_a and t_w will be proportional to the rate of change of t_w .

3. FLOW LOCALISATION

Recent studies [6, 18, 23–25] have considered the problem of flow localisation by introducing a local perturbation in an otherwise uniform specimen and evaluating the subsequent yield behaviour. In all studies the strain rate sensitivity has been assumed to be time invariant and, with the exception of Ref. [6], positive. In the present case it is necessary to include the time dependent nature of the strain rate sensitivity.

Consider a tensile specimen deforming uniformly at constant strain rate. At some point during the test a small portion of the gauge length is assumed to experience small local deviations $\delta\epsilon$, $\delta\dot{\epsilon}$, $\delta\sigma$ and δt_a relative to the uniformly deforming bulk. From equations (2) and (3) assuming a constant tensile load along the test bar

$$M_i \frac{\delta\dot{\epsilon}}{\dot{\epsilon}} = (M_f - M_i) \frac{\delta t_a}{t_a} - \gamma \delta\epsilon \quad (8)$$

where $\gamma = [\partial\sigma/\partial\epsilon]_{t, C_s} - \sigma$ and use has been made of the expression for M_f .

Differentiating equation (8) for $\delta\dot{\epsilon}/\dot{\epsilon} \ll 1$ gives

$$M_i \frac{\delta\ddot{\epsilon}}{\dot{\epsilon}} - (M_f - M_i) \frac{\delta\dot{t}_a}{t_a} + \gamma \delta\dot{\epsilon} = 0 \quad (9)$$

From equation (7)

$$\delta\dot{t}_a = \frac{\delta t_w - \delta t_a}{\tau} \quad (10)$$

where

$$\delta t_w = -t_w \frac{\delta\dot{\epsilon}}{\dot{\epsilon}} \quad (11)$$

Substituting equations (10) and (11) together with δt_a from (8) into equation (9) gives

$$M_i \frac{\delta\ddot{\epsilon}}{\dot{\epsilon}} + \left(\frac{M_f}{\gamma\tau\dot{\epsilon}} + 1\right) \gamma \delta\dot{\epsilon} + \frac{\gamma\delta\epsilon}{\tau} = 0. \quad (12)$$

Equation (12) may be treated as a second order linear differential equation provided the coefficients vary sufficiently slowly over the period of interest to be approximated by suitably averaged values [26].

Table 1. Values of parameters used in analysis

Parameter	Low carbon steel [11]	Au-4.5 Cu [9, 22]
M_i (MPa)	2.81	$0.538 + 0.0228\epsilon - 0.0013\epsilon^2$
M_f (MPa)	$M_i[1 - 0.011(\epsilon^{0.62}/\dot{\epsilon})^{2/3}]$	$M_i[1 - 1.43(\epsilon^{1/2}/\dot{\epsilon})^{2/3}]$
τ (s)	$3.2 \times 10^{-3} \epsilon^{0.62}/\dot{\epsilon}$	$5.8 \times 10^{-3} \epsilon^{0.88}/\dot{\epsilon}$
σ_f (MPa)	$78.5 + 909\epsilon^{0.49}$	$50.1 + 656\epsilon^{0.73}$

Introducing new parameters

$$\phi = \frac{1}{2} \left(\frac{M_f}{\gamma \dot{\epsilon} \tau} + 1 \right) \frac{\gamma \dot{\epsilon}}{M_i} \quad (13)$$

and

$$\lambda = \left(\frac{\gamma \dot{\epsilon}}{M_i \tau} \right)^{1/2} \quad (14)$$

gives the characteristic equation

$$r^2 + 2\phi r + \lambda^2 = 0. \quad (15)$$

For $\phi^2 < \lambda^2$, the characteristic roots are complex, giving

$$\delta \epsilon = C e^{-\phi t} \sin(\theta t + \alpha) \quad (16)$$

where $\theta^2 = \lambda^2 - \phi^2$ and C and α are constants determined from the boundary conditions $\delta \epsilon = \delta \epsilon_0$ and $\delta \dot{\epsilon} = \delta \dot{\epsilon}_0$ at $t = 0$, where t is measured from the time that the perturbation is introduced. The value of $\delta \dot{\epsilon}_0$ may be determined from $\delta \epsilon_0$ using equation (8), since $\delta t_a = 0$ at $t = 0$.

For $\phi^2 > \lambda^2$, the characteristic roots are real, giving

$$\delta \epsilon = e^{-\phi t} [c_1 \exp(-bt) + c_2 \exp(bt)] \quad (17)$$

where c_1 and c_2 are the boundary value constants, and $b = (\phi^2 - \lambda^2)^{1/2}$.

Of importance is the sign of ϕ . If $\phi > 0$, equation (12) is stable and $\delta \epsilon$ approaches zero in the limit of large t . Therefore a small local perturbation $\delta \epsilon_0$ introduced at $t = 0$ will decay with time. On the other hand, if $\phi < 0$ equation (12) is unstable and $\delta \epsilon$ will be a sensitive function of $\delta \epsilon_0$. Any local perturbation will grow in time, resulting in localised yielding. Therefore $\phi = 0$ may be taken as the point of instability where bifurcation to non-uniform deformation occurs. Two cases are evident. If $\phi < 0$ and $\phi^2 < \lambda^2$, $\delta \epsilon$ will oscillate with an exponentially increasing amplitude as given by equation (16). If $\theta < 0$ and $\phi^2 > \lambda^2$, $\delta \epsilon$ will grow exponentially with time [equation (17)].

In Fig. 2 values of M_f , ϕ and θ^2 are plotted as a function of the strain for low carbon steel using the data in Table 1. The values of M_f , ϕ and θ^2 , which are all initially positive, decrease with increasing

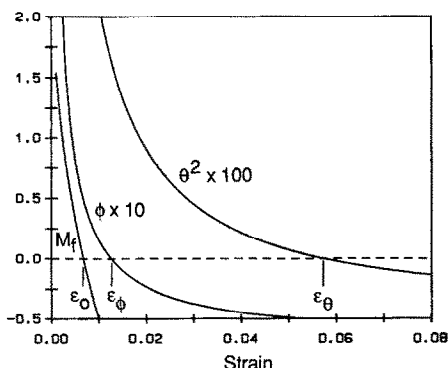


Fig. 2. Effect of strain on M_f , ϕ and θ^2 for Fe-C; $\dot{\epsilon} = 5 \times 10^{-5} \text{ s}^{-1}$.

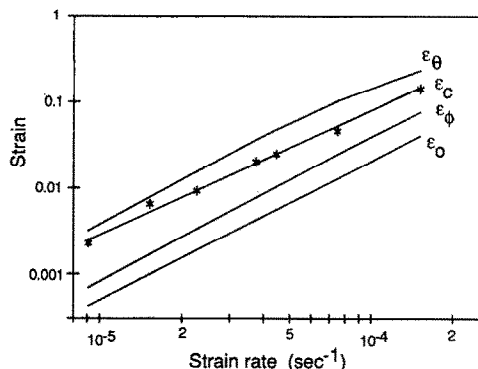


Fig. 3. Effect of strain rate on ϵ_0 , ϵ_ϕ , ϵ_θ and ϵ_c for Fe-C.

strain reaching zero at ϵ_0 , ϵ_ϕ and ϵ_θ respectively, where $\epsilon_0 < \epsilon_\phi < \epsilon_\theta$. At strains between ϵ_ϕ and ϵ_θ , $\phi < 0$ and $\theta^2 > 0$. Therefore, strain localisation is given by equation (16). For strains near ϵ_ϕ strain localisation may develop only slowly due to the small values of ϕ and the sinusoidal variation of $\delta \epsilon$ (and $\delta \dot{\epsilon}$). For $\epsilon > \epsilon_\theta$, $\theta^2 < 0$ and equation (17) is applicable. In this case the strain localisation will occur more rapidly since ϕ is no longer near zero and $\delta \epsilon$ increases directly exponentially with time. It may be argued that $\theta^2 = 0$ provides an upper bound to quasi-stable flow.

Calculations of the effect of applied strain rate on ϵ_0 , ϵ_ϕ and ϵ_θ are shown in Figs 3 and 4 for low carbon steel and Au-4.5% Cu respectively, using the data in Table 1. Also shown are the experimental values of ϵ_c for these two alloys. It is evident that for both alloys the respective strain parameters exhibit similar strain rate dependencies. The values of ϵ_c all lie between ϵ_ϕ and ϵ_θ , indicating that [in agreement with equation (16)] a finite time or strain interval is required for the strain rate to localise into a discernible deformation band. For Au-4.5% Cu the values of ϵ_c lie relatively close to the ϵ_θ boundary. Unfortunately, application of the linear perturbation theory developed in this section is only valid for defining the point of instability and the very early stage of localisation where $\delta \dot{\epsilon}/\dot{\epsilon} \ll 1$. To examine the subsequent growth of the instability into an observ-

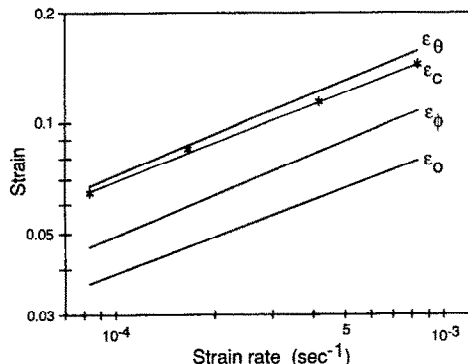


Fig. 4. Effect of strain rate on ϵ_0 , ϵ_ϕ , ϵ_θ and ϵ_c for Au-4.5% Cu.

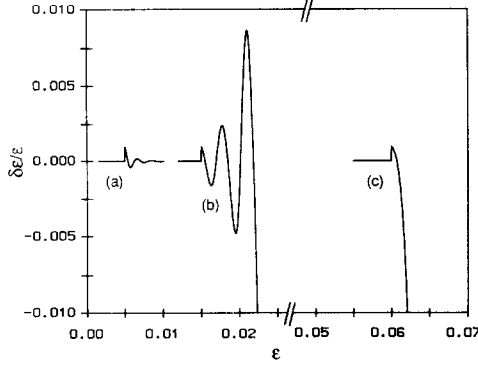


Fig. 5. Effect of local strain perturbation on subsequent yield behaviour; $\dot{\epsilon} = 5 \times 10^{-5} \text{ s}^{-1}$, (a) $\epsilon_x = 0.005$, (b) $\epsilon_x = 0.015$, (c) $\epsilon_x = 0.06$.

able deformation band it is necessary to use numerical methods.

3.1. Numerical simulation

The growth of a local strain inhomogeneity has been modelled by dividing the specimen gauge length into a number of discrete sections. Each section is characterised by local values of ϵ , $\dot{\epsilon}$ and t_a . It is assumed that all sections obey the same constitutive relations [equations (2), (3) and (7)] and interaction between adjacent sections is neglected. The sum of the strain rates of each section is equal to the applied strain rate, which is assumed to be constant

$$\dot{\epsilon} = \frac{1}{N} \sum_{i=1}^N \dot{\epsilon}_i \quad (18)$$

where $\dot{\epsilon}_i$ is the local strain rate of the i th section and N is the number of sections ($N = 10$ in this analysis). At each time step the values of $\dot{\epsilon}_i$ and σ were obtained from iterative solution of equations (2) and (18) using current values of ϵ_i and $t_{a,i}$. The values of ϵ_i were obtained by numerically integrating equation (2) using a fourth order Runge-Kutta method. The values of $t_{a,i}$ were obtained from numerical integration of equation (7) using Euler's method and assuming $\tau = t_w$. A normalised time step, $dt = 2.5 \times 10^{-5} \dot{\epsilon}^{-1}$, was used in all calculations. The

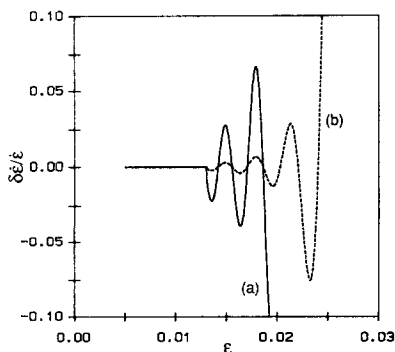


Fig. 6. Effect of $\delta\epsilon_0$ on the magnitude of strain rate oscillations; (a) $\delta\epsilon_0/\epsilon = 10^{-2}$, (b) $\delta\epsilon_0/\epsilon = 10^{-3}$.

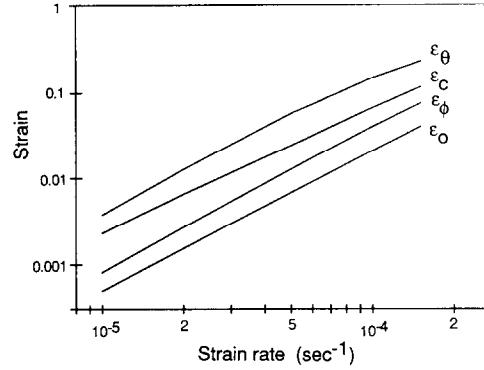


Fig. 7. Effect of strain rate on calculated values of ϵ_c for Fe-C.

calculated values of strain rate were found to be essentially independent of the magnitude of the time step for smaller values of dt . The expressions for M_i , M_f , t_w , and σ_f given in Table 1 for low carbon steel were used in all calculations.

At the start of the calculation all sections were given the same initial strain. On reaching a particular value of strain, ϵ_x , the strain in one section was discontinuously increased by $\delta\epsilon_0/\epsilon = 10^{-3}$ relative to the other sections.

In Fig. 5 the calculated variation of $\delta\epsilon/\epsilon$ is shown for three values of ϵ_x . The values of ϵ_x shown cover the three cases; $\epsilon_0 < \epsilon_x < \epsilon_\phi$, $\epsilon_\phi < \epsilon_x < \epsilon_\theta$ and $\epsilon_\theta < \epsilon_x$. For $\epsilon_x < \epsilon_\phi$ [curve (a)] the strain discontinuity decays sinusoidally as predicted by equation (16) with $\phi > 0$. For curve (b), $\epsilon_\phi < \epsilon_x < \epsilon_\theta$ and the amplitude of $\delta\epsilon$ increases since $\phi < 0$ and $\theta^2 > 0$. For $\epsilon_0 < \epsilon_x$ the strain bifurcates directly without any oscillations in agreement with equation (16).

It is clear from curve (b) of Fig. 5 that for $\epsilon_\phi < \epsilon_x < \epsilon_\theta$ a finite strain interval, $\Delta\epsilon$, is required for the initial strain rate perturbation to grow to the point of being macroscopically observed, i.e. $\epsilon_c = \epsilon_x + \Delta\epsilon$. It is also apparent that $\Delta\epsilon$ will depend on the magnitude of $\delta\epsilon_0$. In Fig. 6 calculated $\delta\epsilon/\epsilon$ curves for $\delta\epsilon_0/\epsilon$ equal to -10^{-3} and -10^{-2} respectively are given. It may be shown from equation (16) that for $\delta\epsilon/\epsilon \ll 1$ the ratio $\delta\epsilon/\delta\epsilon_0$ is constant. Therefore the number of oscillations required for $\delta\epsilon$ to be observed, and hence $\Delta\epsilon$, increases with decreasing $\delta\epsilon_0$.

In Fig. 7, calculated values of ϵ_c for Fe-C with $\delta\epsilon_0/\epsilon = -10^{-3}$ applied at ϵ_ϕ assuming a minimum observable value of $\delta\epsilon/\epsilon = 10^{-1}$ are plotted as a function of strain rate. Also shown are the curves for ϵ_0 , ϵ_ϕ and ϵ_θ . It is seen that the curves all exhibit a similar strain rate dependence and that the calculated values of ϵ_c are in reasonable agreement with the experimental values (Fig. 3).

4. DISCUSSION

It is well established that in the absence of dynamic strain ageing plastic deformation at constant temperature may be characterised by constitutive re-

lations of the form

$$\dot{\epsilon} = f\{\sigma, \rho_i, \rho_j, \dots\} \quad (19)$$

where ϵ and σ are extrinsic state variables and $\rho_i, \rho_j, \dots$ are intrinsic state structure variables which are determined by so called evolutionary equations

$$\rho_i = f_i\{\dot{\epsilon}, \sigma, \rho_i, \rho_j, \dots\} \quad (20)$$

which describe the progressive development of structure related state variables during deformation. The use of a single structure parameter, ρ_f , the Forrest dislocation density, has been shown to describe continuous deformation over a wide range of strains [27]. Recent studies [18, 23] have included the mobile dislocation density as a second structure variable to account for strain localisation at small strains.

To include the effect of dynamic strain ageing, an additional state parameter, C_s , must be included in the constitutive relation. Therefore

$$\dot{\epsilon} = f\{\sigma, \rho_f, \rho_m, C_s\} \quad (21)$$

where C_s is described by an evolutionary equation of the form

$$\dot{C}_s = f\{\dot{\epsilon}, \rho_m, C_s\} \quad (22)$$

as given by equations (3) or (4) and (7). The effective solute composition is essentially a "fast" variable [18], having a relaxation time approximately equal to t_w .

The time dependent character of C_s is fundamental to dynamic strain ageing phenomena. If C_s was an instantaneous function of strain and strain rate, transient effects would not be observed and the strain rate sensitivity would be described by a single parameter, M . In this case $M < 0$ would be a valid criterion for localised yielding [3, 6]. However, this is clearly not the case, and flow localisation is determined by both the local strain hardening rate and the rate of change of C_s .

The results of the numerical simulation confirm that $\phi \leq 0$ is the criterion for the onset of flow localisation due to dynamic strain ageing. The local strain and strain rate oscillations predicted by the linear perturbation theory are shown to increase in amplitude until a deformation band is formed. The increasing amplitude of the $\delta\epsilon$ and $\delta\dot{\epsilon}$ oscillations for $\epsilon_\phi < \epsilon < \epsilon_\theta$ are in striking agreement with the experimental measurements of Wijler and Shade van Westrum [28]. These investigators observed oscillations in local strain and strain rate measured over a small portion of the gauge length during tensile tests on a Au-14%Cu alloy. The amplitude of the oscillations increased with increasing strain such that after approximately three "observable" oscillations the strain rate localised into a deformation band. Numerical simulation of Portevin-Le Chatelier effect subsequent to deformation band formation requires the use of equation (4) rather than (3), and is treated elsewhere [29].

As shown in Figs 3 and 4, ϵ_0 , ϵ_ϕ and ϵ_θ all exhibit

a similar dependence on strain rate. The strain rate dependencies of ϵ_0 , ϵ_ϕ and ϵ_θ are given by

$$\epsilon_0^{m+\beta} = \left(\frac{1}{K_1}\right)^{3/2} \dot{\epsilon} \quad (23)$$

$$\epsilon_\phi^{m+\beta} = \left[\frac{1}{K_1} \left(1 + \frac{\gamma\dot{\epsilon}\tau}{M_i}\right)\right]^{3/2} \dot{\epsilon} \quad (24)$$

$$\epsilon_\theta^{m+\beta} = \left\{ \frac{1}{K_1} \left[1 + \frac{\gamma\dot{\epsilon}\tau}{M_i} \left(1 + 2 \left(\frac{M_i}{\gamma\dot{\epsilon}\tau} \right)^{1/2} \right) \right] \right\}^{3/2} \dot{\epsilon} \quad (25)$$

Comparison of equations (23)–(25) shows that ϵ_0 , ϵ_ϕ and ϵ_θ will exhibit the same dependence on strain rate only if $\gamma\dot{\epsilon}\tau/M_i$ is independent of strain. In general the individual strain dependencies of γ and τ tend to cancel as evidenced by the data in Table 1, resulting in the ratios ϵ_ϕ/ϵ_0 and $\epsilon_\theta/\epsilon_0$ being nearly constant.

Experimental measurements [9, 11, 12] indicate that ϵ_c also exhibits a similar dependence on strain rate. It is evident that ϵ_c may substantially exceed ϵ_ϕ . As discussed previously, ϵ_c is determined by ϵ_ϕ and the strain interval, $\Delta\epsilon$, required for the strain to bifurcate into an observable discontinuity. It is clear that the value of $\Delta\epsilon$ will be a function of the sensitivity of the method of strain measurement employed, and a constant strain resolution must be used to obtain consistent results. In addition, as shown in Fig. 6, $\Delta\epsilon$ is also a function of the magnitude of the local strain or strain rate fluctuations present during testing. Such fluctuations may occur as a result of local structural inhomogeneities and are clearly important in determining $\Delta\epsilon$. In the limit of a uniform, homogeneous specimen macroscopic strain localisation may not be evident until ϵ_θ is reached. Although not specifically treated here, $\Delta\epsilon$ can also be shown to be a function of the uniformity of the cross-sectional area of the specimen. It is evident, therefore, that $\Delta\epsilon$ is a "technique" and "material" sensitive variable. As a consequence, the use of ϵ_c measurements to determine parameters such as $m + \beta$ or the activation energy for serrated flow must be viewed with some caution. Measurements of M_f and ϵ_0 , while more tedious to carry out, allow a more fundamental measure of the important parameters influencing dynamic strain ageing.

5. CONCLUSIONS

1. Flow stress transients which accompany changes in strain rate in the regime of dynamic strain ageing are due to the time dependent change of the effective solute composition at mobile dislocations. The change from the instantaneous strain rate sensitivity to the final quasi-steady state value may be described by a first order relaxation process.

2. Constitutive equations describing plastic flow behaviour associated with dynamic strain ageing must include C_s as an additional state variable.

3. Linear perturbation theory and numerical simulation define an instability strain, ϵ_ϕ , above which local strain inhomogeneities grow with increasing

strain. The value of ϵ_ϕ is greater than the strain required for negative values of M_f .

4. The localisation of strain and strain rate into an identifiable deformation band requires a finite strain interval which is a function of the magnitude of the local strain inhomogeneity and the sensitivity of measurement.

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